

Phase 2

Requirement Analysis Phase

Functional, non-functional, data and environment requirements

1. Introduction

This phase defines what the AI StudyBuddy backend must do, who will use it, what data it handles and what environment it needs. The requirements below are taken from the project's description, features, roles, architecture and setup sections.

2. Stakeholders / User Roles

Role	Responsibilities
Student	Primary user. Register, log in securely, upload study materials, generate summaries, flashcards, quizzes and study plans, view and manage previously generated resources.
Administrator	Manage overall system, monitor registered users, manage user accounts, oversee application usage, maintain system integrity, ensure AI services function correctly, perform system-level management and resolve user issues.

3. Functional Requirements

ID	Requirement	Description
FR-01	User registration and login	Users create an account and log in with email and password; a JWT access token is issued on login.
FR-02	Secure authentication	Protected routes verify the JWT; passwords are hashed with bcrypt before storage.
FR-03	Role-Based Access Control	Students and Administrators have different privileges; admin-only routes are blocked for other roles.
FR-04	Study material management	Authenticated users upload and manage study materials (text content) used as AI input.
FR-05	AI summary generation	Generate a concise summary of an uploaded material using the Gemini API.
FR-06	AI flashcard generation	Generate question-answer flashcards from a material for active recall.
FR-07	AI quiz generation	Generate multiple-choice questions with options and the correct answer.
FR-08	AI study plan generation	Generate a personalized schedule from the learning goal, hours per day and number of days.
FR-09	Persistence and history	Store generated resources in MongoDB so users can retrieve them later.
FR-10	Administrative operations	Admin APIs to manage users and monitor system activity.

4. Non-Functional Requirements

- **Security:** JWT-based authentication, bcrypt password hashing, RBAC, secrets kept in environment variables (.env).
- **Scalability and maintainability:** modular architecture separating routes, controllers, middleware, models and utilities (MVC pattern).

- **Reliability of startup:** the server starts listening only after a successful database connection; a global error handler returns consistent JSON errors.
- **Input validation:** file uploads limited to .txt, .md and .pdf files with a 5 MB size limit.
- **Interoperability:** RESTful JSON APIs consumable by the React frontend; CORS enabled for the client origin.
- **Extensibility:** new AI capabilities can be added with minimal architectural changes.

5. Data Requirements

Entity	Data stored
User	_id, name, email (unique), password (hashed), role (student/admin), createdAt
Study Material	_id, userId, title, subject, content, createdAt
Summary	_id, userId, materialId, summary, createdAt
Flashcard	_id, userId, materialId, question, answer, createdAt
Quiz	_id, userId, materialId, questions (array), createdAt
Study Plan	_id, userId, studyPlan, examDate, createdAt

6. External Interface Requirement

Google Gemini AI API (model: Gemini 2.5 Flash) is used for all AI tasks: summaries, flashcards, quizzes and study plans. A valid GEMINI_API_KEY is required.

7. Software Requirements

- **Operating System:** Windows 10/11, macOS or Linux (cross-platform).
- **Node.js (v16 or above):** runtime for server logic and routing.
- **npm (v8 or above):** package manager for dependencies.
- **Express.js:** lightweight web framework for RESTful entry points.
- **MongoDB:** NoSQL database for users, materials and AI-generated resources.
- **Postman:** API testing tool (Thunder Client can also be used).
- **Code Editor:** Visual Studio Code or similar IDE.

8. Hardware Requirements

Component	Requirement
Processor	Intel Core i5 (8th Gen or above) / AMD Ryzen 5 or equivalent
RAM	8 GB minimum (16 GB recommended for concurrent MongoDB, processing layers and execution nodes)
Storage	1 GB of available disk space

9. Constraints and Assumptions

- Users must be authenticated before using AI features.
- AI output quality and availability depend on the Google Gemini API and a valid API key.
- MongoDB must be running and environment variables configured before the server starts.
- Only text-based study content is used as AI input in the current implementation.